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# Dynamic Multi-Level Heuristic Framework for Predicting Lung Cancer from CT Scan Data

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**ABSTRACT:** Lung cancer is the abnormal growth of cells in the lung, causing a severe risk to human health because the lung has a connected network of blood vessels and lymphatic channels vulnerable to metastasis. Lung cancer is known to be one of the most dangerous diseases, which is the main reason for disease and death when diagnosed in primitive stages. Since lung cancer can only be detected more broadly after it spreads to the lung region and the occurrence of lung cancer in the earlier stage is very difficult to predict. It causes a greater risk as radiologists and specialist doctors assess the existence of lung cancer. The accuracy of detecting tumor shape, location, and volume plays a vital role in the successful diagnosis and treatment of tumors. For this reason, it is important to build a smart and automatic cancer prediction system that is accurate and at which stage of cancer, or improve the accuracy of the previous cancer prediction, which will help determine the type of treatment and treatment depth depending on the severity of the disease. Adaptive Hierarchical Heuristic Mathematical Model (AHIHMM) has been proposed for the deep learning approach. To analyze deep learning based on the historical therapy scheme in the development of Non-Small Cell Lung Cancers (NSCLC), automated radiation adaptation protocols for optimizing local tumor regulation at lower rates. Computed Tomography (CT) images are widely used for lung cancer detection as they provide information about different parts of the lung. Furthermore, the proposed system consists of three steps, including acquiring the image, preprocessing, segmentation, and classification. Firstly preprocessing stage, (i) Improved Median Filter (IMF) and Wiener Filter (WF) techniques are introduced for removing or reducing the noise from the image, (ii) Image histogram methods are used for image quality enhancement, Histogram Rehabilitated (HR) and Local Contrast Enhancement (LCE) techniques that consists of two stages of processing to increase the potentiality of contrast enhancement and to preserve the local details in the images, contrasted details to improve the quality of the CT pulmonary image, then (iii) resulted image passed to edge detection done using Canny Edge Detection (CED) algorithm. Secondly segmentation stage, the segmentation of the region injured by cancer by the use of the K-means clustering algorithm, using the enhanced lung CT image. Segmentation of the lung CT image is carried out to extract any significant feature of a segmented image, and a specific feature extraction method is implemented. Finally, classification stage, identification of lung cancer through an explosion-trained Deep Learning Neural Network (DLNN). The image training should be carried out utilizing a deep-learning NN before the classification process is done. The experimental results show that metrics such as accuracy and Efficiency Ratio. The experimental results have been demonstrated with the help of a dataset downloaded from online. The experiment result shows that the proposed AHIHMM achieves the highest accuracy compared with the other modern methods, such as Deep Convolutional Neural Network (DCNN) and Artificial Neural Network (ANN).

**KEYWORDS:** Adaptive HIerarchical Heuristic Mathematical Model (AHIHMM), Lung Cancer Prediction (LCP), Computed Tomography (CT), Improved Median Filter (IMF), Deep Learning Neural Network (DLNN).

## 1 Introduction

Cancer is one of the leading causes of death worldwide. It has caused more deaths than any other disease, such as tuberculosis or malaria. World Health Organization (WHO) agencies for cancer research (i.e, International Agency for Cancer Research (IACR) and American Cancer Society) report that 17.1 million new cancer cases were recorded in 2018 worldwide [1]. WHO estimates that cancer incidences might increase to 27.5 million by 2040, with an estimated 16.3 million deaths expected as a result of cancer [1].

A malignant tumor labeled by uncontrolled cell growth in lung tissues is lung cancer, known as lung carcinoma. This must be treated so that its growth is not spread to other parts of the body, resulting in metastases. Carcinoma is the majority of cancers starting in the lung. The two main types are carcinoma of small cells of the lung and non-small cells of the lung. The primary factor in 88% of lung cancers is long-term smoking. Approximately 15-20 % of cases occur in humans who have not smoked except for asthma, smoking, asbestos, and radon gas [2].

Lung cancer arises when cells in the lung grow uncontrollably, forming a tumor lesion or nodule. The reasons behind lung cancer are smoking, air pollution, harmful chemicals, etc. Based on the cancer type, stage, and molecular characteristics, treatment includes surgery, radiation therapy, chemotherapy, immunotherapy, and targeted therapy. Computed Tomography (CT) scan is one of the widely used medical imaging technologies that presents useful information for various medical and clinical diagnoses. Practical manual segmentation from CT images used to measure the volume of lung tumors, which is operator-dependent and time-consuming [3].

The first and fundamental step for lung image analysis is the segmentation of the lung. Segmentation is the process by which regions of interest of a digital image are portioned and labeled in different segments using properties such as intensity, texture, etc., which is much easier to analyze. Errors in lung segmentation would generate false information, so accurate segmentation is necessary. The segmentation of lung tumors is very heuristic in nature as the CT images include bones, bronchioles, trachea, soft tissues, and background artifacts, which share properties of intensity, connectivity with tumor lesions [4].

A CT scan combines a series of X-ray images taken from different angles and uses computer processing to create cross-sectional image slices of the bones, blood vessels, and soft tissues. Special x-ray equipment is used for a CT scan of the chest to examine abnormalities in the lung [5].

The traditional method to predict the presence of lung cancer is CT and X-rays. Biopsy, which is usually performed with a bronchoscope or a CT scan, confirms the diagnosis. The impact of death due to cancer in men is due primarily to lung cancer, and a new and comprehensive approach to the early diagnosis of lung cancer should, therefore, be established [6]. The method of image recognition has been enhanced by recent technology in deep learning or Deep Neural Networks (DNN). Scan the pattern in an image utilizing DNN and decide if it recognizes the pattern. Besides, it looks for several patterns when analyzing an image. Neural network training requires a predetermined dataset to be used by the network to recognize, learn, and classify a picture. DNN has become more and more common because it can easily be used for recognizing image patterns and classifying images [7]. Most of the patients with Non-Small-Cell Lung Cancer (NSCLC) are ineligible because of locally improved diseases or remote metastases and therefore radiation treatment.

In the proposed work, DNN can be used in various medical fields to extract features and classify images. The DNN model has been trained and tested using functional CT data. The images have been divided into one convolutional layer only. It is difficult to classify heavier images with DNN. While images of the form CT are used primarily in medical imaging, unnecessary artifacts can be created. The threshold has been used to prevent the images from displaying these objects. The threshold procedure eliminates unwanted peaks in pixels of the medical image of lung cancer. The main contributions of the proposed work steps are:

- Pre-processing stage, IMF and WF techniques are introduced for removing or reducing the noise from the image, HR and LCE techniques to increase the potentiality of contrast enhancement and to preserve the local details in the images, contrasted details to improve the quality of the CT pulmonary image, and then edge detection is done using the CED algorithm.
- Adaptive HIerarchical Heuristic Mathematical Model (AHIHMM) to predict lung cancer based on Deep learning using CT images.
- DNN framework to extract high-level features directly from the information and apply the K-means clustering algorithm.
- The experimental results have been performed to enhance the accuracy in prediction, and the dataset http://diagnijmegen.nl/index.php/Lung_Cancer has been utilized.

The rest of the paper is organized as follows: The Literature review on LCP existing schemes is briefly discussed in section 2. The proposed methodology for the LCP system using a deep learning algorithm is presented in section 3. The results and discussion are briefly explained in section 4. The conclusion is described in section 5.

## 2 Literature Review

Coudray et al [8] developed a Deep Convolutional Neural Network (inception v3) on whole-slide images obtained from The Cancer Genome Atlas to accurately and automatically classify them into LUAD, LUSC, or normal lung tissue. The performance of the method is comparable to that of pathologists, with an average Area Under the Curve (AUC) of 0.97. The proposed model was validated on independent datasets of frozen tissues, formalin-fixed paraffin-embedded tissues, and biopsies. Furthermore, trained the network to predict the ten most commonly mutated genes in LUAD. Also, found that six of them-STK11, EGFR, FAT1, SETBP1, KRAS, and TP53-can be predicted from pathology images, with AUCs from 0.733 to 0.856 as measured on a held-out population. These findings suggest that deep-learning models can assist pathologists in the detection of cancer subtypes or gene mutations.

Wang et al [9] proposed a Support Vector Machine (SVM) for the prediction of malignant and benign lung tumors. The early diagnosis of lung cancer plays a very important role in therapeutic regimen selection. However, lung cancers are spatially and temporally heterogeneous; this limits the use of invasive biopsy. But radiomics, which refers to the comprehensive quantification of tumour phenotypes by applying a large number of quantitative image features, can capture intra-tumoral heterogeneity in a non-invasive way. Here, carry out a radiomic analysis of 150 features quantifying lung tumour image intensity, shape, and texture. These features are extracted from 593 patients' Computed Tomography (CT) data on the Lung Image Database Consortium Image Database Resource Initiative (LIDC-IDRI) dataset. By using SVM, we find that a large number of quantitative radiomic features have diagnostic power. The accuracy of the prediction of malignant lung tumors is 86% in the training set and 76.1% in the testing set. As CT imaging of lung tumors is widely used in routine clinical practice, a radiomic classifier will be a valuable tool that can help clinical doctors diagnose lung cancer.

Jakimovski and Davcev [10] introduced a double Convolutional Deep Neural Network (CDNN) for lung cancer stage detection in Computed Tomography (CT) images. CT scans to train a CDNN and a regular CDNN. These topologies were tested against lung cancer images to determine the Tx cancer stage at which these topologies can detect the possibility of lung cancer. The first step was to pre-classify the CT images from the initial dataset so that the training of the CDNN could be focused. Next, built the double CDNN with max pooling to perform a more thorough search. Finally, used CT scans of different Tx cancer stages of lung cancer to determine the Tx stage in which the CDNN would detect the possibility of lung cancer. Then tested the regular CDNN against the double CDNN. Using this algorithm, doctors will have additional help in early lung cancer detection and early treatment. After extensive training with 100 epochs, the double CDNN obtained the highest accuracy, whereas the regular CDNN obtained only lower accuracy.

Teramoto et al [11] introduced a Deep Convolutional Neural Network (DCNN) for an automated classification scheme of lung cancers. An automated classification scheme for lung cancers presented in microscopic images using a DCNN, which is a major deep learning technique. DCNN used for classification consists of three convolutional layers, three pooling layers, and two fully connected layers. In evaluation experiments conducted, the DCNN was trained using the original database with a graphics processing unit. Microscopic images were first cropped and resampled to obtain images with a resolution of 256×256 pixels and, to prevent overfitting, collected images were augmented via rotation, flipping, and filtering. The probabilities of three types of cancers were estimated using the developed scheme, and their classification accuracy was evaluated using threefold cross-validation. In the results obtained, approximately 71% of the images were classified correctly, which is on par with the accuracy of cytotechnologists and pathologists. Thus, the developed scheme is useful for the classification of lung cancers from microscopic images.

Yu et al [12] developed an Adaptive HIerarchical Heuristic Mathematical Model (AHIHMM) for lung cancer prediction. To analyze deep learning based on the historical therapy scheme in the development of Non-Small Cell Lung Cancers (NSCLC), automated radiation adaptation protocols that aim at optimizing local tumor regulation at lower rates of grade 2 RP2 radiation pneumonitis. Furthermore, the system proposed consists of several steps, including acquiring the image, preprocessing, binarization, thresholding, and segmentation, extraction of features, and detection of a Deep Neural Network (DNN). Segmentation of the lung CT image is carried out to extract any significant feature of a segmented image, and a specific feature extraction method is implemented. The test evaluation showed that the model proposed could detect the highest accuracy of the absence or presence of lung cancer.

## 3 Proposed Methodology

AHIHMM has been proposed for the deep learning approach. CT images are widely used for lung cancer detection as they provide information about different parts of the lung. Furthermore, the proposed system consists of three steps, including acquiring the image, preprocessing, segmentation, and classification. Firstly preprocessing stage, (i) IMF and WF techniques are introduced for removing or reducing the noise from the image, resulted while keeping its features intact for better understanding and recognition (ii) Image histogram methods are HR and LCE techniques that consists of two stages of processing to increase the potentiality of contrast enhancement and to preserve the local details in the images, contrasted details to improve the quality of the CT pulmonary image, resulted image improve quality enhancement (iii) resulted image passed to edge detection done using CED algorithm, the most useful image technique to improve quality of the image analysis process, result in the blurring of fine-scaled image edges. Secondly segmentation stage, the segmentation of the region injured by cancer by the use of the K-means clustering algorithm, using the enhanced lung CT image. Segmentation of the lung CT image is carried out to extract any significant feature of a segmented image, and a specific feature extraction method is implemented. Finally, classification stage, identification of lung cancer through an explosion-trained DLNN. The image training should be carried out utilizing a deep-learning NN before the classification process is done.

In a wide variable list, the Bayesian network technique has been used for graph theory to define the radiotherapy environment that allows the hierarchical links between variables and interesting results to be recognized. The method used is based on the identification from a high-dimensional data set of separate extended Markov blankets (MBs) for RP2 and LC.

***Case 1: Deep Neural Network (DNN)*** Let's consider the data $\{y_j\}$with labels $\{z_j\}$such that $\chi = \{(y_j, x_j) | y_j \epsilon \Re^m, z_j \in \Re^n,\ j = 1, \dots M,$ a DNN determines a function $f_{DNN}: \Re^m \to \Re^n$ to weave via data such that $f_{DNN}(y_j) \cong z_j$ as much as possible through the utility of 3 different parts: Neurons $x_j \in \Re$, layers of l neurons $x = (x_1, .., x_l)$ and activation function $\rho$, if a deep neural network has layers $i = 0, \dots k,$ each of which has $m_i$ neurons, then $i = 0$ and $k$ would indicate the initial input and last output layer, respectively. An activation function $\rho: \Re^{m_{i-1}} \to \Re^{m_i}$ links the neurons of (i-1)th layer $x^{(i-1)} \in \Re^{m_{i-1}}$ would satisfy:

$$\mathrm{x}^{(\mathrm{i})} = \rho\left(\Theta^{(\mathrm{i}-1)} . x^{(i-1)} + a^{(i-1)}\right) \quad (1)$$

A typical option of $\rho$ is a Rectified Linear Unit (ReLU) or sigmoid. Where $\Theta^{(\mathrm{i}-1)} \in \Re^{m_i \times m_{i-1}}$ and $a^{(i-1)} \in \Re^{m_{i-1}}$ denotes the unknown weights and bias to be calculated.

### *3.1 Image-Preprocessing*

Medical image processing methods have been used to predict changes present in the lungs of the recorded CT pulmonary images. The lung cancer processing structure is shown in Figure 1. A lung tumor has been successfully detected by the medical image processing technique according to the processing structure of the lung picture. Image de-noising is an asset for image noise processing, which includes filtering techniques that include different ways to de-noise an image. Accordingly, noises are spotted with neighboring information and are removed using the best filtering techniques without affecting the image quality, and reinforce the smoothness of the image taken for examination. Image preprocessing is important to remove the noise and to enhance the quality of the image. IMF is used to remove the salt and pepper noise, and WF is useful to remove speckle noise. HR and LCE algorithms are used to increase the contrast of the image.

#### *3.1.1 Median Filter (MF)*

MF is a very popular technique for the removal of impulse noise because it runs through the signal cell by cell and replaces the value of each cell with the neighboring by median of the intensity levels of its mathematical accuracy. Due to this tendency of MF, the good pixel cells are replaced by the corrupted ones. Therefore, de-noising often leads to the removal of fine details present in an image because it is done at the cost of distorted and blurred features possessed by this filtering technique. The noisy value of the digital image or the sequence is replaced by the median value of the neighborhood (mask). The median filtering output is $\mathrm{g(x, y) = med\{ f\,(x - i, y - j), i, j \in W\}}$ where $\mathrm{f\,(x, y), g(x, y)}$are the original image and the output image respectively, W is the two-dimensional mask: the mask size is $\mathrm{n \times n}$ (where n is commonly odd) such as $3 \times 3, 5\ \times 5$, etc.; the mask shape may be linear, square, circular, cross, etc. Because the median filter is nonlinear, its mathematical analysis is relatively complex for an image with random noise.

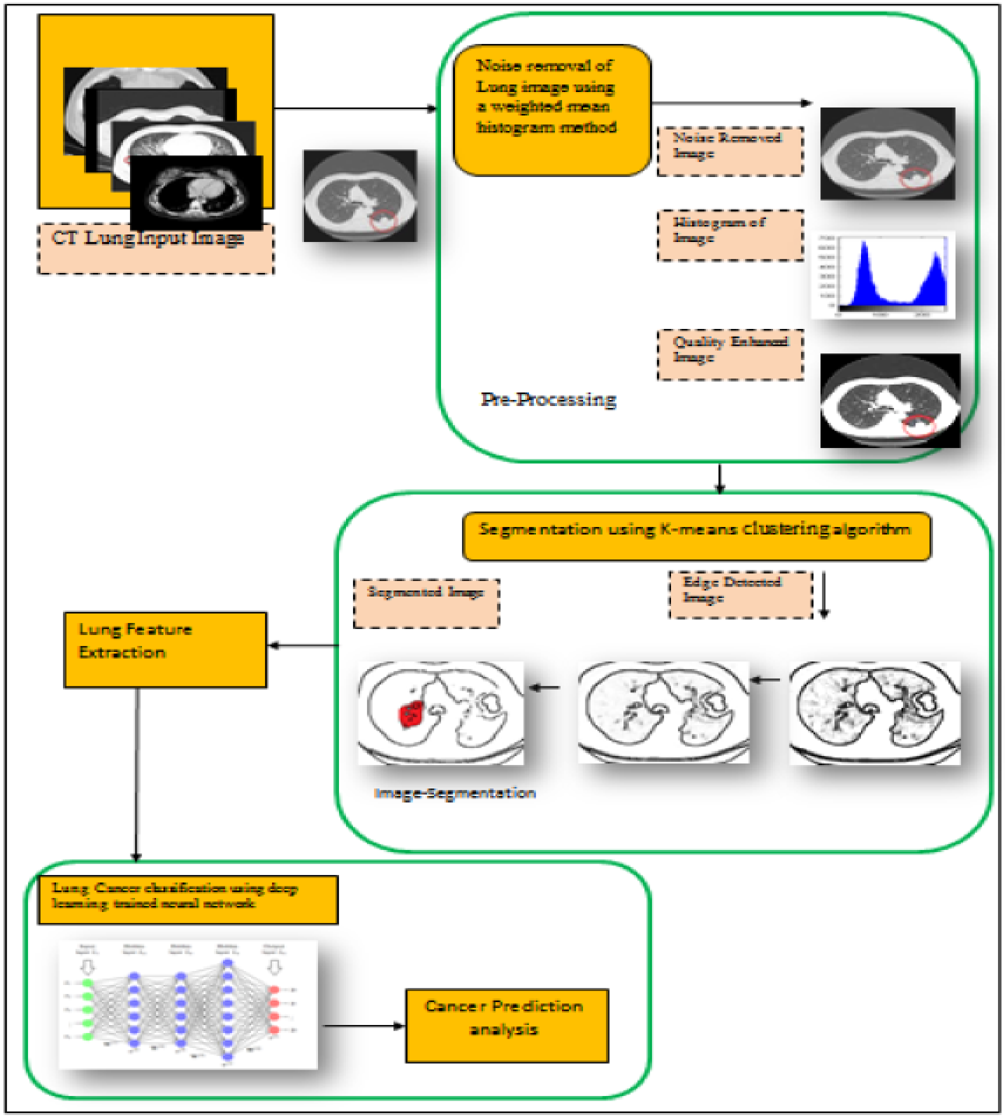


**Figure 1:** The Architecture of the Proposed AHIHMM Method

For an image with zero mean noise under a normal distribution, the noise variance of the median filtering is approximately [13].

$$\sigma_{med}^2 = \frac{1}{4nf^2(\bar{n})} \approx \frac{\sigma_i^2}{n+\frac{\pi}{2}-1} \cdot \frac{\pi}{2} \tag{2}$$

where $\sigma_i^2$ is input noise power (the variance), n is the size of the median filtering mask, $f\ (\bar{n})$ is the function of the noise density. And the noise variance of the average filtering is,

$$\sigma_0^2 = \frac{1}{n}\sigma_i^2 \tag{3}$$

The median filtering performance of random noise reduction is better than the average filtering performance, but for the impulse noise, especially when narrow pulses are farther apart, and the pulse width is less than $\frac{n}{2}$, the median filter is very effective. The median filter's noise-reducing effects depend on the size and shape of the filtering mask, and its algorithm complexity mainly depends on how to get the median value. In order to improve the noise-reducing performance of the median filter, improved methods for the Improved Median Filter (IFM) are proposed.

*3.1.2 Improved Median Filtering (IFM)*

The filtering mask is mainly an $\mathrm{n} \times \mathrm{n}$ square mask or a cross mask. The smaller the mask is, the better the image details are retained, the weaker the noise reduction performance is; the larger the mask is, the less the image details are retained, the stronger the noise reduction performance. In the filtering process, it can adaptively resize the mask according to the noise levels of the mask. In the mask, $max$ is the maximum value of gray levels, $min$ is the minimum value of gray levels, $average$ is the average value of gray levels, $med$ is the median value of gray levels, $f(i,j)$ is the central value of the mask, $n$ is the size of the mask. The adaptive filtering requires two steps:

Step 1: adaptively resizing the mask

Initialization: let $n = 3$;

Computation: $A1 = med - min, A2 = med - max$

Judgment: if $A1 > 0$ and $2 < 0$, then turn to step 2; if not, then enlarge the size of the mask, let $n = n + 2$.

Step 2: median filtering.

The average filter has better performance for filtering random noise. Combining the median filter with the average filter for a certain size of the filtering mask. For the natural image, neighboring pixels have a strong correlation. The gray value of each pixel is quite close to neighboring pixels, and the edge pixels also have the same property. In the reducing-noise process, sequentially check each pixel. If the value of a pixel is greater than the average value in the mask, then judge that the pixel is contaminated by the noise and replace it with the median value of the mask; otherwise, retain the original value of the pixel unchanged. This forms an iterative process; it not only decreases the time complexity but also improves the noise-reducing effect.

*3.1.3 Weiner Filter (WF)*

The Weiner Filter (WF) is used to filter out the noise present in an image that possesses a corrupted signal in it. This filtering technique uses a statistical approach to filter the noise from each pixel of an image. WF is a technique that performs optimal trading involving opposite filtering and noise smoothing. It removes the blurring and additional noise present in the image, and it is also very optimal in relation to the mean squared error, where it minimizes the overall Mean Square Error in the operation of the filtering technique for noise removal. WF is usually defined by the following [14]:

- Hypothesis: additive noise and image signal are inactive linear random processes containing spectral characteristics.
- Necessity: The filter must be able to achieve and be accessed.

Performance criteria: It depends on the minimum Mean Squared Error.

*3.1.4 Contrast Enhancement*

Histogram Equalization (HE) uniformly distributes the output histogram by using the cumulated histogram as its mapping function. HE has large backward difference values of mapping functions, and the contrast enhancement potential should be enriched without losing the fine details in the CT image. It is important to note that when the input histogram distribution is already uniform, the mapping obtained from cumulating the distribution is $C(r_k) = r_k$, $r_k$ is the kth gray level in the image that identically maps input to output [15].

*Histogram Rehabilitation (HR):* The main objective of this method is to find a rehabilitated histogram $\overline{rg}$ that is closer to a uniform histogram, and to make the difference between $(\overline{rg} - hg_i)$ rehabilitated and input histogram small, which in turn increases the potential of image contrast enhancement, and the resultant image would be more relevant to the input image. The optimization problem for the lower sub image can be defined as

$$\min\|\overline{rg} - hg_i\| + \varsigma_1\|\overline{rg} - u_{iLower}\| \tag{4}$$

The rehabilitated histogram for the lower sub image can be obtained by finding an analytical solution as follows:

$$(\overline{rg}_{Lower} - hg_{iLower}) + \varsigma_1\overline{rg}_{Lower} - \varsigma_1 u_{iLower} = 0 \tag{5}$$

where $\overline{rg}_{Lower}$, $hg_{iLower}$, $u_{iLower}$, and $\varsigma_1$ are the rehabilitated histogram, input histogram, uniform histogram, and the contrast enhancement parameter for the lower sub histogram, respectively.

Then, the bicriteria optimization problem for the upper sub image can be written as,

$$\min\left\|\overline{rg}_{Upper} - hg_{iUpper}\right\| + \varsigma_2\left\|\overline{rg}_{Upper} - u_{iUpper}\right\| \tag{6}$$

The rehabilitated histogram for upper sub image can be obtained by finding an analytical solution as follows:

$$\left(\overline{rg}_{Upper} - hg_{iUpper}\right) + \varsigma_1\overline{rg}_{Upper} - \varsigma_2 u_{iUpper} = 0 \tag{7}$$

where $\overline{rg}_{Upper}$, $hg_{iUpper}$, $u_{iUpper}$, and $\varsigma_2$ are the rehabilitated histogram, input histogram, uniform histogram, and the contrast enhancement parameter for the lower sub histogram, respectively.

$$u_{iLower}, u_{iUpper} \in u_i \forall\ u_i \in R^{256\times 1} \tag{8}$$

*Local Contrast Enhancement (LCE):* Although the global approach for image contrast enhancement is suitable for some cases. The number of pixels in this area may have a negligible influence on the computation of the global transformation. The histogram modification process is the first step in the given CT image, and then the local contrast enhancement method is applied to histogram histogram-rehabilitated image. Let $\mathbb{d}r$ denote a discrete random variable representing discrete gray levels in the range [0, L−1] and let $P(\mathbb{d}r_i)$ is the normalized histogram component corresponding to the $i$th value of $\mathbb{d}r$. The $n$th moment of $\mathbb{d}r$ about its multipeak mean $(\mathfrak{m}\mu)$ for the purpose of enhancing the contrast as well as preserving essential details for any given gray scale and colour images.is defined as,

$$\mathfrak{m}\mu_n(\mathbb{d}r) = \sum_{i=0}^{L-1}\left(\mathbb{d}r_i - \sum_{i=0}^{NP}\frac{P(\mathbb{d}r_i)}{NP}\right)^n P(\mathbb{d}r_i) \tag{9}$$

The local contrast enhancement for the CT image is obtained by applying the following equation after the histogram-rehabilitated image is obtained.

$$g(x,y) = \begin{cases} E.f(x,y) & if\ \mathfrak{m}\mu_{x,y} \le K_0 M \\ f(x,y) & otherwise \end{cases} \tag{10}$$

where $g(x,y)$ and $f(x,y)$ are final enhanced and histogram rehabilitated images, respectively, E, and $K_0 <\ 1$ are specified parameters. $M$ is the global multipeak mean of the input image applied to the LCE method. These two parameters afford a simple yet powerful local contrast enhancement by varying the enhancement parameter in the histogram modification process.

*3.1.5 Canny Edge Detection (CED)*

Edge detection is a fundamental tool in image processing, machine vision, and computer vision, particularly in the areas of feature detection and feature extraction. CED is a technique to extract useful structural information from different vision objects and dramatically reduces the amount of data to be processed. Canny has found that the requirements for the application of edge detection on diverse vision systems are relatively similar [16]. The general criteria for edge detection include:

1. Detection of edges with a low error rate, which means that the detection should accurately catch as many edges shown in the image as possible.
2. The edge point detected by the operator should accurately localize on the center of the edge.
3. A given edge in the image should only be marked once, and where possible, image noise should not create false edges.

The Canny algorithm, as the first step, applies the LoG filter, which emphasizes the region of fastest intensity change. LoG value is used to decompose the original image into two images. After applying the LoG filter follows the usual steps: image smoothing and filtering, finding the gradient magnitude and gradient direction, non-maxima suppression, and edge tracking by hysteresis [17].

***Process of the CED Algorithm***

The process of the CED algorithm can be broken down into 5 different steps:

1. Apply the LoG filter to smooth the image
2. Find the intensity gradients of the image
3. Apply non-maximum suppression to get rid of the spurious response to edge detection
4. Apply a double threshold to determine potential edges
5. Detection of edges by suppressing all the other edges that are weak and not connected to strong edges.

***3.3 Image-Segmentation***

Segmentation of the region injured by cancer by the use of the K-means clustering algorithm using the enhanced lung CT image. The implemented segmentation approach inspects the pixel similarity in the lung CT images and divides the images into several sub-settings to predict the area of concern. The pixel or information is subsequently tested during the image fragmentation process to predict the comparability of data using the same value of spectrum or pixels. The pixel similarity attribute involves the quantitative image evaluation used to accurately shape the cluster. A modified K-means clustering algorithm has been proposed. In the data of lung cancer modified K-means clustering algorithm has been utilized, and the datasets have been divided into two groups of clusters: one cluster of lung cancer patients and the other for cancer-free patients. Based on the value of the attributes, the number of clusters is determined. Two clusters are formed based on these plotted points and two centroid points. The effect of the new attributes on the cluster is described as if the next attribute is smoking, the cluster moves to either left or right to a certain distance, depending on the cluster's impact. K-means clustering is a data-partitioning algorithm that iteratively assigns n observations to exactly one of the k clusters defined by centroids, where k is chosen before the algorithm starts.

Let $X = \{x_1, x_2, \dots, x_n\}$ be the set of n data points and $C = \{c_1, c_2, \dots, c_n\}$ be the set of m cluster centers. The $n$ data points are to be clustered into $m$ cluster centers.

$$J = \sum_{i=1}^{m} \sum_{j=1}^{n} \left\| x_i - c_j \right\|^2 \tag{11}$$

Where $k =$ number of clusters, $n =$ total number of data, $c_j = j^{th}$ cluster center and $x_i = i^{th}$ data point. The optimal centers are obtained by minimizing this objective function.

### *3.4 Image-Classification*

The identification of lung cancer through an explosion-trained Deep Learning Neural Network (DLNN). The image training should be carried out utilizing a deep-learning NN before the classification process is done, since it does not need manual features. Alternatively, the deep learning process uses a captured segmented image or lung CT image that utilizes a huge number of hidden layers to identify the boundaries of the image and the corresponding features utilized for the network's creation, with a large amount of information.

*Case 2: Generative Adversarial Nets (GANs)* have been utilized to synthesize more radiotherapy-like information, to mitigate the issue of small sample size in datasets when modeling difficult state transitions in the circumstances of radiation. A GAN contains two neural networks, one generative (H). The other is biased (E), which seeks to test the (dis)-similarity of the condensed data against the actual data. In general, H is designed to generate inextricable data to confuse E, while E tries to differentiate between the data produced by H or not. The adversary E and H character are generated using the loss function of the mini-max two players:

$$\min_{H} \max_{E} L(H, E) = \min_{H} \max_{E} G_{y \sim Q_{data(y)}}[\log E(y)] + G_{x \sim Q_{prior(x)}}[\log \left(1 - E\left(H_{(x)}\right)\right)] \tag{12}$$

#### *3.4.1 Deep Q Networks*

In Reinforcement Learning (RL), there is a Markov Decision Process (MDP) and an agent. RL has been utilized to mimic how doctors decide on the dose fraction tradeoff required to dictate to a patient. Over a decision made by the present state $\pi: W \to B$ an agent acquires respective reward T and gets reinforced to the next state. An agent taking care of supplying tasks $b \in B$ in a condition, which is explained by the different states $w \in W$ in the conditions. A second opinion or the presence of a specialist in the setting would be an external agent. In the next phase of a dose-escalation radiotherapy therapy course, the required dose per fraction (adaptation) will be determined.

$$P^{\pi}(w, b) = G\left[\sum_{l=1}^{\infty} \delta^k T\left(w_l, \pi(w_l)\right) | \, \pi, w_0 = w, b_0 = b\right] \tag{13}$$

The convergence meets at the single fixed point as $j \to \infty$ based on the contraction mapping theorem,

$$\tilde{P}^*(w, b) = G_{r \sim Q_{wb}}[T(w, b) + \delta \max_{a \in B} \tilde{P}^*(r, a)] \tag{14}$$

*Case 3: Radiotherapy Dose Adaptation* is a key objective of has to determine whether it can potentially use reinforcement learning to regenerate previously made established decisions before attempting to use Deep Reinforcement Learning for autonomous decision-making. A therapy has total fractions $f(r)$ fraction/dose $b(r)$ as a function of time r with V:

$$f(r_*) = f_0, f(R) = 30, b(r) = \begin{cases} b_0, & 0 \le r \le r_* \\ b_1, & r_* \le r \le R \end{cases} \tag{15}$$

The rest of the dose is given for the target, depending on the avid mid-therapy defined through the Metabolic Tumor Volume (MTV). Firstly, local discovery algorithms based on constraints using the Markov blankets have been used to select high-dimensional (297 candidates) variables that relate mostly to RP2/LC. Then, to build a single Bayesian network from the attributes chosen for both the RP2 and the LC, with graphic learning algorithms. Finally, cross-validation has been used for determining and assessing nodes in a Bayesian Network with the highest Area Under Curve value of joint prediction for LC and RP2. This procedure resulted in 9 selected features, including SNPs, cytokines, PET, and miRNA, which have been deemed to reflect the state variables as important predictors.

Deep Neural Network structure and weights $\Theta$ that should be finetuned for the best estimation of the actual Q-function:

$$P_{\mathrm{DNN}}^{\Theta} : \mathrm{W} \rightarrow \mathrm{T}^{|\mathrm{B}|} \; by P_{\mathrm{DNN}}^{\Theta}(w) = \begin{pmatrix} P(w, b_1) \\ P(w, b_2) \\ \vdots \\ P(w, b_l) \end{pmatrix} \tag{16}$$

## 4 Results and Discussion

The findings of segmentation included nodules connecting to the inner wall and precise segmentation of the lung parenchyma. In the classification and detection of objects, efficient segmentation is very important. An efficient method of segmentation is developed based on the distribution of lung CT images by spatial pixel intensity. CT images will simply remove pulmonary parenchyma in morphological operations; however, tissue around the lung can be included. The experimental results have been performed to enhance the accuracy in prediction and the dataset http://diagnijmegen.nl/index.php/Lung_Cancer has been utilized. The experimental results show that metrics such as accuracy and Efficiency Ratio. The experiment result shows that the proposed AHIHMM compares with the other modern methods, such as DCNN and ANN.

### *4.1 Accuracy*

Accuracy in low-dose CT images, and the same approaches are recommended for separating the lungs into high-resolution CT images. The measured value is compared with the extracted and trained features for cancer classification based on deep learning. Double-time matching enhances the accuracy of prediction and decreases the failure rate efficiently. The mean accuracy of the classifier specifies how many samples have been correctly predicted by the total number of samples.

$$Accuracy = \frac{True\ Positive + True\ Negative}{Totalno\ of\ predictions} * 100 \tag{17}$$

### *4.2 Efficiency Ratio*

The efficiency of the proposed AHIHMM method was evaluated in various aspects of recognition rate, misclassification ratio. The high number of CT data has enabled us to use a moderate mini-batch size and to regulate training processes to achieve adequate speed and efficiency. In general, manual inspection is difficult to identify these quantitative image features, computerized methods can efficiently identify those features, and our classifiers can be used effectively in routine practices.

**Table 1:** Performance comparison metrics vs. LCP classification methods.

| Methods | Metrics | |
|---|---|---|
| | Accuracy (%) | Efficiency Ratio (%) |
| **DCNN** | 97.8 | 92.7 |

| | | |
|---|---|---|
| **ANN** | 98.2 | 95.3 |
| **AHIHMM** | 99.5 | 97.5 |

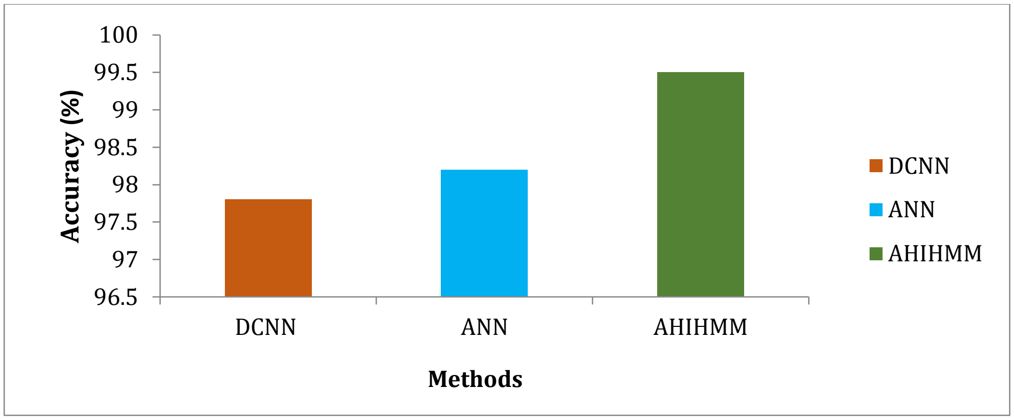


**Figure 2:** Accuracy results evaluation of LCP classification methods

The accuracy results comparison of the three classification methods is shown in Figure 2. The results disclose that the proposed AHIHMM classifier provides higher accuracy results of 99.5%, whereas other existing methods, such as DCNN and ANN, give 97.8% and 98.2% respectively.

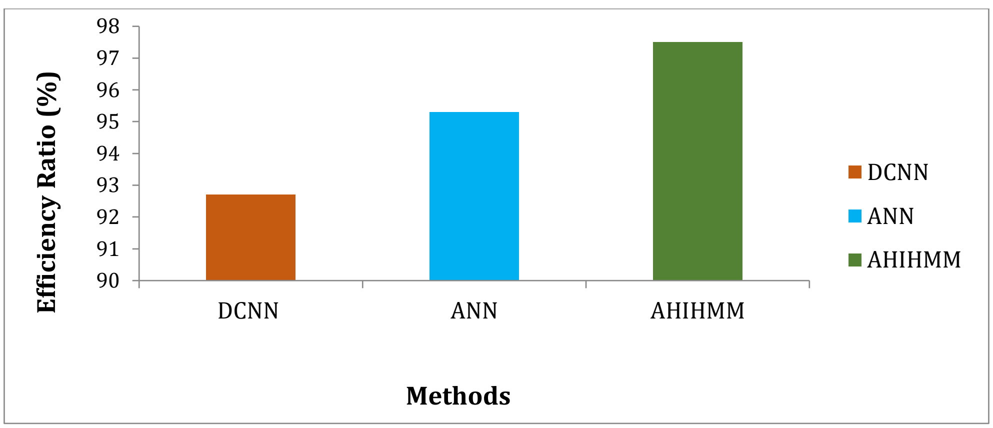


**Figure 3:** Efficiency ratio results evaluation of LCP classification methods

Figure 3 shows the efficiency ratio comparison results of the three classification methods. Those methods are DCNN, ANN, and AHIHMM. The results disclose that the proposed AHIHMM classifier provides a higher efficiency ratio of 97.5%, whereas other existing methods, such as DCNN, ANN, give of 92.7% and 95.3% respectively.

## 5 Conclusion

Deep learning assisted AHIHMM to predict lung cancer on CT images, preprocessing stage, IMF and WF for removing or reducing the noise from the image, HR and LCE techniques to increase the potentiality of contrast enhancement and to preserve the local details in the images for improve the quality, then uses Modified K-means algorithm to pre-classify pictures into slices of images in the same image, where the DNN will concentrate on the image classification of images in similar images.

The next thing is the convolution layer with filtered edges to scan for lung cancer thoroughly. Then, estimating the weighted mean function, which replaces the pixel utilizing the cumulative distribution and likelihood distribution method, improved the image's quality. The injured portion has been segmented by a pixel-like value measurement after the image has been improved. Based on the similarity calculation, spectrally related features have been extracted. The AHIHMM system predicts computed tomography scanning images of lung cancer successfully.

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